

D785

DevOps Security

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A CI pipeline's service account has full administrator rights on the entire cloud account, but the pipeline only deploys to one storage bucket and one container service. The account was created by a former platform lead. What is the best change?

- A. Re-create the account under the current platform lead's name.
- B. Require multi-factor sign-in for human administrators.
- C. Keep admin rights but rotate the account's key every month.
- D. Limit the account to only the permissions needed on that bucket and that container service.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A developer committed a database password to a private repository shared by 25 engineers. She noticed within an hour and removed it in a new commit. The repository is hosted on a well-known service. What must happen next?

- A. Rotate the password and move it into a secrets manager.
- B. Move the repository to a different hosting service.
- C. Nothing more; private repositories are always safe for secrets.
- D. Nothing more; the new commit removed it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

An application needs an API key at runtime. Which approach best keeps the key out of source code?

- A. Retrieve it at runtime from a secrets manager with access limited to that application.
- B. Hard-code it, encoded in Base64.
- C. Turn on disk encryption for the source-code server.
- D. Put it in a configuration file committed to the repository.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Any of 30 developers can push directly to the main branch, which deploys automatically to production. The team ships about 40 changes a week. Which control best keeps an unreviewed change from reaching production?

- A. Let authors approve their own pull requests.
- B. Cut the weekly change count to 10.
- C. Protect main so merges need another developer's approval and passing checks.
- D. Log every push and review the logs each quarter.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A team of 12, split across two offices, wants to find known vulnerabilities in the open-source libraries its web app imports. Which type of tool fits best?

- A. Dynamic application security testing (DAST).
- B. Software composition analysis (SCA).
- C. An annual penetration test, which catches every library flaw.
- D. Static application security testing (SAST).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A scan runs against the deployed test site, sending requests and watching the responses, with no access to source code. Which type of testing is this?

- A. Peer code review.
- B. SCA.
- C. SAST.
- D. DAST.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Two findings are open. Finding A: likelihood 4, impact 5, on the internet-facing login page. Finding B: likelihood 2, impact 5, on an internal reporting tool. Finding B was reported by an external auditor. The team can fix one this sprint. Using risk = likelihood × impact, which should it fix first?

- A. Finding A, risk 20 versus 10.
- B. Finding B, because lower likelihood means higher risk.
- C. Finding B, because an external auditor reported it.
- D. Either one; both have impact 5, so risk is equal.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A risk has a score of 12 and an impact of 4, where risk = likelihood × impact. What is its likelihood?

- A. 8
- B. 3
- C. 12
- D. 48

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

An artifact repository stores build packages on disk in a data center with badge-controlled doors. Build agents download the packages over the internal network. An audit finds anyone capturing network traffic can read the packages. The data center passed a physical security review last month. Which control addresses the finding?

- A. Encrypt the repository's disks at rest.
- B. Skip encryption, since internal networks are always trusted.
- C. Encrypt downloads in transit with TLS.
- D. Add a second badge reader at the data center door.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Over a quarter, four security incidents were detected 2, 6, 10, and 22 hours after they began. Each was contained 3 hours after detection. The security team has 6 analysts. What was the mean time to detect?

- A. 8 hours
- B. 10 hours
- C. 6.7 hours
- D. 13 hours

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A team's mean time to detect was 10 hours across 5 incidents. Four of them took 4, 6, 8, and 12 hours to detect. How long did the fifth take?

- A. 40 hours
- B. 10 hours
- C. 20 hours
- D. 7.5 hours

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Monitoring shows a build server sending data to an unknown outside address at 2 a.m. The server was installed by a vendor and runs a popular CI tool. The on-call engineer has confirmed it is a real incident. What is the next step?

- A. Write the lessons-learned report.
- B. Wait for the vendor's analysis before taking any action.
- C. Wipe and reinstall the server right away.
- D. Contain it by isolating the server from the network while preserving evidence.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A CI pipeline's service account has full administrator rights on the entire cloud account, but the pipeline only deploys to one storage bucket and one container service. ~~The account was created by a former platform lead.~~ What is the best change?

Target word: right-size pipeline permissions **Decides:** The pipeline needs two resources, not the whole account.

A	Camouflage	Ownership is not the issue; scope is.
B	Adjacent category	Protects human logins, not the automation account's scope.
C	Dropped element	Rotation helps, but the excess rights remain.
D	Correct	Least privilege shrinks what a compromised pipeline can reach.

2. Answer A Concept 2

A developer committed a database password to a private repository shared by 25 engineers. She noticed within an hour and removed it in a new commit. ~~The repository is hosted on a well-known service.~~ What must happen next?

Target word: response to a leaked secret **Decides:** The secret remains in history and clones.

A	Correct	Once exposed, a secret must be replaced; storage must change so it does not recur.
B	Camouflage	The host did not cause the leak, and moving keeps the history.
C	Absolute language	Everyone with access, and any leak of the repo, exposes it.
D	Dropped element	Earlier commits and every clone still hold the password.

3. Answer A Concept 2 · transfer

An application needs an API key at runtime. Which approach best keeps the key out of source code?

Target word: safe storage of a runtime secret **Decides:** The key must be delivered at runtime, not stored in code.

A	Correct	The key is stored centrally, controlled, and rotatable.
B	Confusion set	Encoding is not encryption; anyone can decode it.
C	Adjacent category	Anyone with repository access still reads the key.
D	Inverted operation	That places the secret in source control.

4. Answer C Concept 3

Any of 30 developers can push directly to the main branch, which deploys automatically to production. ~~The team ships about 40 changes a week.~~ Which control best keeps an unreviewed change from reaching production?

Target word: preventive control on main **Decides:** Direct pushes skip any second person.

A	Dropped element	Drops the second person who makes review meaningful.
B	Camouflage	Fewer changes can still be unreviewed.
C	Correct	Separation of duties and required checks block unreviewed changes.
D	Adjacent category	A detective control; the change is already in production.

5. Answer B Concept 4

A team of 12, ~~split across two offices~~, wants to find known vulnerabilities in the open-source libraries its web app imports. Which type of tool fits best?

Target word: tool for third-party library flaws **Decides:** The target is dependencies, not the team's own code.

A	Adjacent category	DAST probes the running app, not the dependency list.
B	Correct	SCA inventories dependencies and matches them to known vulnerabilities.
C	Absolute language	Point-in-time tests miss many flaws and new disclosures.
D	Confusion set	SAST reads the team's own source code patterns.

6. Answer D Concept 4 • transfer

A scan runs against the deployed test site, sending requests and watching the responses, with no access to source code. Which type of testing is this?

Target word: name the tool from its behavior **Decides:** Tests the running app from outside.

A	Adjacent category	A manual review of source, not a runtime scan.
B	Confusion set	SCA inspects the component list, not live behavior.
C	Inverted operation	SAST analyzes code without running it, the reverse of this.
D	Correct	Dynamic testing exercises the running application.

7. Answer A Concept 5

Two findings are open. Finding A: likelihood 4, impact 5, on the internet-facing login page. Finding B: likelihood 2, impact 5, on an internal reporting tool. ~~Finding B was reported by an external auditor.~~ The team can fix one this sprint. Using risk = likelihood × impact, which should it fix first?

Target word: higher-risk finding **Decides:** A scores 20; B scores 10.

A	Correct	$4 \times 5 = 20$ beats $2 \times 5 = 10$, and A is internet-facing.
B	Inverted operation	Risk rises with likelihood.
C	Camouflage	The reporter does not change the risk.
D	Dropped element	Drops likelihood from the formula.

8. Answer B Concept 5 • transfer

A risk has a score of 12 and an impact of 4, where risk = likelihood × impact. What is its likelihood?

Target word: solve for likelihood **Decides:** Likelihood = risk / impact.

A	Confusion set	Subtracts, treating the score as additive.
B	Correct	$12 / 4 = 3$.
C	Wrong stem number	The risk score itself.
D	Inverted operation	Multiplies instead of dividing.

9. Answer C Concept 6

An artifact repository stores build packages on disk in a data center with badge-controlled doors. Build agents download the packages over the internal network. An audit finds anyone capturing network traffic can read the packages. ~~The data center passed a physical security review last month.~~ Which control addresses the finding?

Target word: control for exposure on the network **Decides:** The exposure is in transit.

A	Confusion set	Protects stored data, not packets on the wire.
B	Absolute language	Internal traffic can be captured, as the audit showed.
C	Correct	TLS protects data moving across the network.
D	Camouflage	Physical access was not the finding.

10. Answer B Concept 7

Over a quarter, four security incidents were detected 2, 6, 10, and 22 hours after they began. Each was contained 3 hours after detection. ~~The security team has 6 analysts.~~ What was the mean time to detect?

Target word: MTTD **Decides:** Total detection time divided by incidents.

A	Confusion set	That is the median, not the mean.
B	Correct	$(2 + 6 + 10 + 22) / 4 = 40 / 4 = 10$.
C	Camouflage	Divides by analysts instead of incidents.
D	Adjacent category	Adds containment time, which is a different metric.

11. Answer C Concept 7 · transfer

A team's mean time to detect was 10 hours across 5 incidents. Four of them took 4, 6, 8, and 12 hours to detect. How long did the fifth take?

Target word: solve for the missing value **Decides:** Total = $10 \times 5 = 50$ hours.

A	Inverted operation	Subtracts the mean from the total instead of the four known values.
B	Wrong stem number	The mean itself.
C	Correct	$50 - (4 + 6 + 8 + 12) = 50 - 30 = 20$.
D	Dropped element	Averages the four known values and drops the fifth.

12. Answer D Concept 8

Monitoring shows a build server sending data to an unknown outside address at 2 a.m. ~~The server was installed by a vendor and runs a popular CI tool.~~ The on-call engineer has confirmed it is a real incident. What is the next step?

Target word: next incident-response step **Decides:** Confirmed incident, still active.

A	Confusion set	A real phase, but it comes last.
B	Camouflage	Who installed it does not delay containment.
C	Dropped element	Skips containment and destroys evidence needed to find the cause.
D	Correct	Containment stops the damage; evidence supports analysis and eradication.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Least privilege for pipelines	Service accounts and pipelines get only the permissions their tasks need, scoped to specific resources.	Keeping broad rights and adding other controls around them.	An automation account with admin rights and a narrow job.
2	Secrets management	Secrets never live in code; a leaked secret must be rotated, because version history and copies keep it.	Believing deleting the secret in a later commit fixes the leak.	A password or key found in a repository.
3	Branch protection and separation of duties	Changes to protected branches need review by someone other than the author plus passing checks before merge.	Relying on after-the-fact logging or self-approval.	Direct pushes to a branch that deploys automatically.
4	SAST vs. DAST vs. SCA	SAST reads source without running it; DAST tests the running app from outside; SCA checks third-party components for known flaws.	Mixing up which tool looks at code, runtime, or dependencies.	What the scan looks at: our code, the live app, or libraries.
5	Risk-based vulnerability prioritization	Fix first what has the highest risk, where risk = likelihood × impact, informed by exposure.	Ranking by who reported it or by impact alone.	Two findings with scores and limited capacity.
6	Encryption at rest vs. in transit	At-rest encryption protects stored data; in-transit encryption (TLS) protects data moving over a network.	Applying one to solve the other's problem.	Where the exposure happens: on disk or on the wire.
7	Monitoring metrics (MTTD)	Mean time to detect = total time from incident start to detection ÷ number of incidents.	Adding containment time or dividing by staff count.	A list of detection delays.
8	Incident response order	After confirming an incident, contain it (and preserve evidence) before eradication, recovery, and lessons learned.	Wiping and rebuilding first, destroying evidence.	A confirmed incident and a question about the next step.

Course scope (public description): In the DevOps Security course, students will explore tools to continuously monitor for security risks. By utilizing industry best practices, learners fortify code repositories through robust access control, encryption, monitoring, and vulnerability management strategies. Students will learn about secure access control mechanisms vital for safeguarding build and deployment environments against unauthorized access, data breaches, and malicious activities. Students will analyze advanced techniques to enhance monitoring and reporting processes, empowering effective risk management and informed decision-making. They will gain the techniques needed to quickly identify and resolve incidents, ensuring uninterrupted service and minimizing business impact.